

Partial Differentiation

Notations for Partial Derivatives If $z = f(x, y)$, we write

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Rule for Finding Partial Derivatives of $z = f(x, y)$

1. To find f_x , regard y as a constant and differentiate $f(x, y)$ with respect to x .
2. To find f_y , regard x as a constant and differentiate $f(x, y)$ with respect to y .

EXAMPLE 1 If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ and $f_y(2, 1)$.

SOLUTION Holding y constant and differentiating with respect to x , we get

$$f_x(x, y) = 3x^2 + 2xy^3$$

and so

$$f_x(2, 1) = 3 \cdot 2^2 + 2 \cdot 2 \cdot 1^3 = 16$$

Holding x constant and differentiating with respect to y , we get

$$f_y(x, y) = 3x^2y^2 - 4y$$

$$f_y(2, 1) = 3 \cdot 2^2 \cdot 1^2 - 4 \cdot 1 = 8$$

► **Example 1** Find $f_x(1, 3)$ and $f_y(1, 3)$ for the function $f(x, y) = 2x^3y^2 + 2y + 4x$.

Solution. Since

$$f_x(x, 3) = \frac{d}{dx}[f(x, 3)] = \frac{d}{dx}[18x^3 + 4x + 6] = 54x^2 + 4$$

we have $f_x(1, 3) = 54 + 4 = 58$. Also, since

$$f_y(1, y) = \frac{d}{dy}[f(1, y)] = \frac{d}{dy}[2y^2 + 2y + 4] = 4y + 2$$

we have $f_y(1, 3) = 4(3) + 2 = 14$. ◀

EXAMPLE 4 If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

SOLUTION Using the Chain Rule for functions of one variable, we have

$$\frac{\partial f}{\partial x} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y}$$

$$\frac{\partial f}{\partial y} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = -\cos\left(\frac{x}{1+y}\right) \cdot \frac{x}{(1+y)^2}$$

► **Example 3** Find $\partial z/\partial x$ and $\partial z/\partial y$ if $z = x^4 \sin(xy^3)$.

Solution.

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial}{\partial x}[x^4 \sin(xy^3)] = x^4 \frac{\partial}{\partial x}[\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial x}(x^4) \\ &= x^4 \cos(xy^3) \cdot y^3 + \sin(xy^3) \cdot 4x^3 = x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3) \\ \frac{\partial z}{\partial y} &= \frac{\partial}{\partial y}[x^4 \sin(xy^3)] = x^4 \frac{\partial}{\partial y}[\sin(xy^3)] + \sin(xy^3) \cdot \frac{\partial}{\partial y}(x^4) \\ &= x^4 \cos(xy^3) \cdot 3xy^2 + \sin(xy^3) \cdot 0 = 3x^5 y^2 \cos(xy^3) \quad \blacktriangleleft\end{aligned}$$

EXAMPLE 7 Find the second partial derivatives of

$$f(x, y) = x^3 + x^2 y^3 - 2y^2$$

SOLUTION In Example 1 we found that

$$f_x(x, y) = 3x^2 + 2xy^3 \qquad f_y(x, y) = 3x^2 y^2 - 4y$$

Therefore

$$\begin{aligned}f_{xx} &= \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3 & f_{xy} &= \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2 \\ f_{yx} &= \frac{\partial}{\partial x}(3x^2 y^2 - 4y) = 6xy^2 & f_{yy} &= \frac{\partial}{\partial y}(3x^2 y^2 - 4y) = 6x^2 y - 4 \quad \blacksquare\end{aligned}$$

EXAMPLE 8 Calculate f_{xyz} if $f(x, y, z) = \sin(3x + yz)$.

SOLUTION

$$\begin{aligned}f_x &= 3 \cos(3x + yz) \\ f_{xx} &= -9 \sin(3x + yz) \\ f_{xy} &= -9z \cos(3x + yz) \\ f_{xyz} &= -9 \cos(3x + yz) + 9yz \sin(3x + yz) \quad \blacksquare\end{aligned}$$

► **Example 5** Let $f(x, y) = x^2 y + 5y^3$.

- (a) Find the slope of the surface $z = f(x, y)$ in the x -direction at the point $(1, -2)$.
- (b) Find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(1, -2)$.

Solution (a). Differentiating f with respect to x with y held fixed yields

$$f_x(x, y) = 2xy$$

Thus, the slope in the x -direction is $f_x(1, -2) = -4$; that is, z is decreasing at the rate of 4 units per unit increase in x .

Solution (b). Differentiating f with respect to y with x held fixed yields

$$f_y(x, y) = x^2 + 15y^2$$

Thus, the slope in the y -direction is $f_y(1, -2) = 61$; that is, z is increasing at the rate of 61 units per unit increase in y . ◀

► **Example 12** Find the second-order partial derivatives of $f(x, y) = x^2y^3 + x^4y$.

Solution. We have

$$\frac{\partial f}{\partial x} = 2xy^3 + 4x^3y \quad \text{and} \quad \frac{\partial f}{\partial y} = 3x^2y^2 + x^4$$

so that

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (2xy^3 + 4x^3y) = 2y^3 + 12x^2y$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (3x^2y^2 + x^4) = 6x^2y$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (3x^2y^2 + x^4) = 6xy^2 + 4x^3$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (2xy^3 + 4x^3y) = 6xy^2 + 4x^3 \quad \blacktriangleleft$$

Third-order, fourth-order, and higher-order partial derivatives can be obtained by successive differentiation. Some possibilities are

$$\frac{\partial^3 f}{\partial x^3} = \frac{\partial}{\partial x} \left(\frac{\partial^2 f}{\partial x^2} \right) = f_{xxx}$$

$$\frac{\partial^4 f}{\partial y^4} = \frac{\partial}{\partial y} \left(\frac{\partial^3 f}{\partial y^3} \right) = f_{yyyy}$$

$$\frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial^2 f}{\partial y \partial x} \right) = f_{xyy}$$

$$\frac{\partial^4 f}{\partial y^2 \partial x^2} = \frac{\partial}{\partial y} \left(\frac{\partial^3 f}{\partial y \partial x^2} \right) = f_{xxyy}$$

► **Example 13** Let $f(x, y) = y^2e^x + y$. Find f_{xyy} .

Solution.

$$f_{xyy} = \frac{\partial^3 f}{\partial y^2 \partial x} = \frac{\partial^2}{\partial y^2} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2}{\partial y^2} (y^2e^x) = \frac{\partial}{\partial y} (2ye^x) = 2e^x \quad \blacktriangleleft$$

The wave equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

describes the motion of a waveform, which could be an ocean wave, a sound wave, a light wave, or a wave traveling along a vibrating string. For instance, if $u(x, t)$ represents the displacement of a vibrating violin string at time t and at a distance x from one end of the string (as in Figure 8), then $u(x, t)$ satisfies the wave equation. Here the constant a depends on the density of the string and on the tension in the string.

EXAMPLE 10 Verify that the function $u(x, t) = \sin(x - at)$ satisfies the wave equation.

SOLUTION $u_x = \cos(x - at) \quad u_t = -a \cos(x - at)$

$$u_{xx} = -\sin(x - at) \quad u_{tt} = -a^2 \sin(x - at) = a^2 u_{xx}$$

So u satisfies the wave equation. ■

► **Example 14** Show that the function $u(x, t) = \sin(x - ct)$ is a solution of Equation (6).

Solution. We have

$$\begin{aligned}\frac{\partial u}{\partial x} &= \cos(x - ct), & \frac{\partial^2 u}{\partial x^2} &= -\sin(x - ct) \\ \frac{\partial u}{\partial t} &= -c \cos(x - ct), & \frac{\partial^2 u}{\partial t^2} &= -c^2 \sin(x - ct)\end{aligned}$$

Thus, $u(x, t)$ satisfies (6). ◀

Practice Questions

- Let $f(x, y) = 3x^3y^2$. Find
 - $f_x(x, y)$
 - $f_y(x, y)$
 - $f_x(1, y)$
 - $f_x(x, 1)$
 - $f_y(1, y)$
 - $f_y(x, 1)$
 - $f_x(1, 2)$
 - $f_y(1, 2)$
- Let $z = e^{2x} \sin y$. Find
 - $\partial z / \partial x$
 - $\partial z / \partial y$
 - $\partial z / \partial x|_{(0, y)}$
 - $\partial z / \partial x|_{(x, 0)}$
 - $\partial z / \partial y|_{(0, y)}$
 - $\partial z / \partial y|_{(x, 0)}$
 - $\partial z / \partial x|_{(\ln 2, 0)}$
 - $\partial z / \partial y|_{(\ln 2, 0)}$

3–10 Evaluate the indicated partial derivatives. ■

- $z = 9x^2y - 3x^5y$; $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$
- $f(x, y) = 10x^2y^4 - 6xy^2 + 10x^2$; $f_x(x, y), f_y(x, y)$
- $z = (x^2 + 5x - 2y)^8$; $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$
- $f(x, y) = \frac{1}{xy^2 - x^2y}$; $f_x(x, y), f_y(x, y)$
- $\frac{\partial}{\partial p}(e^{-7p/q}), \frac{\partial}{\partial q}(e^{-7p/q})$
- $\frac{\partial}{\partial x}(xe^{\sqrt{15xy}}), \frac{\partial}{\partial y}(xe^{\sqrt{15xy}})$
- $z = \sin(5x^3y + 7xy^2)$; $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$
- $f(x, y) = \cos(2xy^2 - 3x^2y^2)$; $f_x(x, y), f_y(x, y)$
- Let $f(x, y) = \sqrt{3x + 2y}$.
 - Find the slope of the surface $z = f(x, y)$ in the x -direction at the point $(4, 2)$.
 - Find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(4, 2)$.
- Let $f(x, y) = xe^{-y} + 5y$.
 - Find the slope of the surface $z = f(x, y)$ in the x -direction at the point $(3, 0)$.
 - Find the slope of the surface $z = f(x, y)$ in the y -direction at the point $(3, 0)$.
- Let $z = \sin(y^2 - 4x)$.
 - Find the rate of change of z with respect to x at the point $(2, 1)$ with y held fixed.
 - Find the rate of change of z with respect to y at the point $(2, 1)$ with x held fixed.